

First step: Inform the workshop operation content, inform the operation purpose, and sign the work ticket.



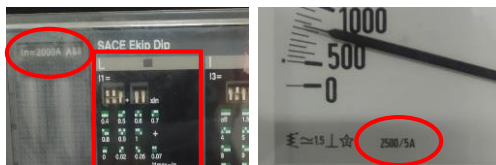
Second step: Prepare the document, Megger Sverker 900, an Electrical Integrity Check Sticker, and a tools kit.



Third step: Choose the appropriate PPE (protective clothing, protective mask, arc flash-proof gloves, earplugs). PPE is required to be tested and within the validity period.



Fourth step: Check the EFR specification, current rating of main breaker (ACB) and CT. Record in the table on the form.



 PT.MILA - SERANG		SETTING & VERIF					
		EARTH FAULT RELAY (EFR) MAIN					
NO	STATION	EARTH FAULT RELAY			MCB RATING	CURRENT TRANSFORMER	
		BRAND	MODEL	RANGE (A)		4CT	2CT
1							
2							
3							
4							
5							
6							
7							

Fifth step: Determine the EFR setting value and operating time by calculating using the formula listed below the form. Record in the table on the form.

Notes :

- 1) EFR Setting (%) : 10-15%
- 2) EFR Setting (Amp) : MCB Rating x (10 ~ 15%)/(Ratio CT)
- 3) Operating Time : Max 0.5 Second

SETTING & VERIFICATION								
EARTH FAULT RELAY (EFR) MAIN DISTRIBUTION BOARD								
RELAY	MCCB RATING	CURRENT TRANSFORMER		SETTING VALUES			TEST RESULT	
RANGE (A)		4CT	2CT	%	Delay (s)	I _n (A)	Actual Trip	Remarks
							I _{sc}	I _{sc}

Sixth step: Open the cover of the EFR section of the MCC cabinet.



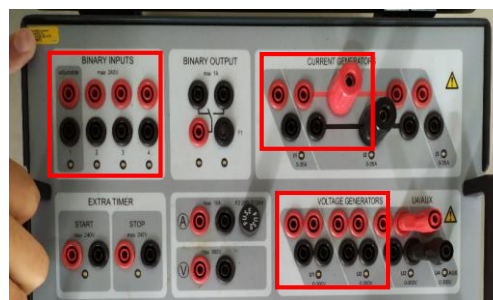
Seventh step: Confirm the breaker (MCB) used as power for the EFR and turn it off.



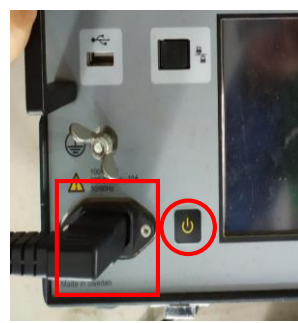
Eighth step: Disconnect all cables which connected to the EFR and secure the cables.



Ninth step: Connect the cables from the voltage generators terminal to the EFR power terminal, the current generators to the EFR current, and the binary inputs to the EFR trip contact.

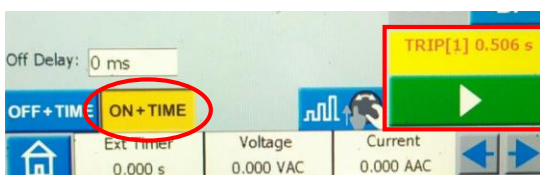


Tenth step: Connect the power Megger Sverker cables to the socket outlet and turn it on.



Eleventh step: Test the EFR Operating time.

Touch ON+TIME and start (touch play symbol). If the value (seen on the display with the words trip[1]) is not equal to approximately 0.5 seconds, turn the time selector switch on the EFR.



Twelfth step: Test the EFR current.

Set the current on Megger Sverker according to the calculation in step five by touching the box next to I1 and touching the play symbol. If the EFR trips, turn the current selector switch on the EFR towards the higher current. If the EFR does not trip, turn the current selector switch on the EFR to a lower current. Note: after each test, press the reset button on the EFR.



Twelfth step: Check the current setting and operating time according to the calculation. Install the green electrical integrity for "Passed" and red for "Not Passed" or in trouble.

